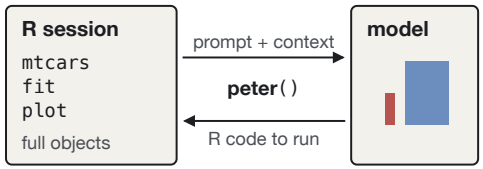
Agents in your R session with gptr :: CHEATSHEET

`peter()` runs language model agents in your current R session, on objects already in memory. gptr reads as a grammar of Peter, after Peter Naur and Peter Cathcart Wason. In the spirit of Naur's readable language definitions, gptr records sessions as runnable documents; Wason's reasoning research with Jonathan Evans inspires its System 1 and System 2 roles.

How it works



R uses full objects; models see descriptions, previews and tool output, including data values (`?gptr_egress`).

gptr_describe(`mtcars`)
What a model is shown of an object.

peter(`"Say hello.",`
`  .opts = list(context = "none"))`
Workspace/object descriptions off; prompts, instructions and environment still go. Remote consent still applies.

Set up

OpenAI for live calls; Try it offline needs no key and defines `fake` and `s` for page 2.

install.packages(`"gptr"`)
library(`gptr`)
From CRAN; R 4.2+.

pak::pak(`"Broccolito/gptr"`)
Development version from GitHub; install pak first if needed.

gptr_login(`"openai"`)
Interactive R: enter your OpenAI key, hidden. Kept for future sessions; `scope = "session"` or `"keyring"` keeps it elsewhere.

gptr_providers(); **gptr_models**(`"gpt"`)
List the routes; search the offline model catalog.

gptr_config(`model = "gpt",`
`  .scope = "user"`)
Save a default model (session, project or user).

gptr_init(`"."`)
Create `.gptr/`; keep sessions and records.

gptr_trust(`".", trust = TRUE`)
Trust the project's settings, skills and plugins.

`e = ` **gptr_config**()`$egress`
`e$openai = "ack"`
gptr_config(`egress = e, .scope = "user"`)
Consent to OpenAI context in scripts.

Try it offline

```
step = list(tool = "r", input = list(
  code = "fit = lm(mpg ~ wt, mtcars)"))
fake = gptr_fake_provider(
  list(step, "Done."))
s = peter("Fit mpg on wt.", mtcars,
  model = fake, mode = "auto"); coef(fit)
```

After `library(gptr)`: no key or network. The scripted replies fit the model, then set `s$text` to "Done."

Ask Peter from R

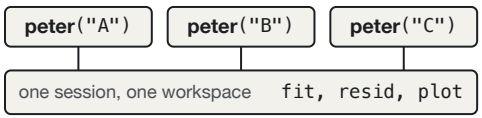
Live calls follow Set up. `mtcars` is built in; quoted model names are strings. In functions use `envir = new.env()` to keep objects.

`fit_s = ` **peter**(`"Fit mpg ~ wt as fit.",`
`  mtcars, mode = "auto"`)
Prompt, then data. `auto` permits ordinary changes in scripts; critical actions still ask.

fit_s$text; fit_s$value; fit_s$turns
Answer, newest `gptr_return()` value (else NULL), prompt count.

peter(`"Summarize.", mtcars,`
`  budget = list(tokens = 5e4, turns = 5)`)
Per-call limits; `turns` counts model requests.

Pipe: one session, many turns



`fit_s |> ` **peter**(`"Plot the residuals."`)
Continue the live fit above, with its workspace.

`fit_s |> ` **peter**(`"Check the fit.",`
`  model = "gpt:high"`)
Change reasoning; keep the history.

`b = ` **gptr_fork**(`fit_s`)
`b |> ` **peter**(`"Try a log model."`)
Branch; reference-object edits can reach the original.

`fit_s; ` **summary**(`fit_s`)`; ` **gptr_last**()
Answer, history, most recent session.

The console

gptr_doc(`"log.R"`)
`session = ` **peter**()
`session |> ` **peter**()
Interactive R: bind a transcript, then chat. `/exit` returns the session; the next call reopens it.

INPUT SYNTAX AT `peter>`

`Fit mpg on wt.`	A prompt for the session
`!head(mtcars)`	Run R and show the model
`!!x = 2`	Run R; send no code or output
```` ```r … ``` ````	Fenced R, run like `!code`
`""" … """`	A multi-line prompt
`line \`	Continue on the next line
`@notes.md`	Your existing file: first 40 lines
`@fit`	Attach a workspace object
`{pi}`	Insert an R value

SLASH COMMANDS: ONE AT A TIME

`/help [cmd]`	Commands and syntax
`/model /mode`	Show or switch either
`/status`	Session, model, mode
`/cost /env`	Tokens and cost; objects
`/tools`	Tools and helpers
`/compact`	Compact the history
`/clear`	New chat, same objects
`/resume /fork`	Resume or fork a session
`/doc [path]`	The history document
`/skills /mcp`	List skills, MCP servers
`/<name>`	A `.gptr/prompts/` template
`/permissions`	Show or change rules
`/undo /redo`	Undo or redo a turn
`/retry`	Resend the last prompt
`/rewind [k]`	Keep turns 1 to k
`/exit`	Leave; return the session

WHILE AN ANSWER RUNS

In a terminal (not Windows) type, then Enter steers and Tab follows up. Ctrl-C (RStudio, R.app: Esc) pauses: **[s]**teer, **[f]**ollow-up, **[c]**ontinue, **[a]**bort, **[b]**ackground. Approvals: **[y]**es, **[a]**lways (session), **[p]** always (project), **[n]**o (`n <text>` says what to do instead).

System 1 and System 2

System 1 typed decision → `TRUE "lung" 1.88` vector + probabilities

System 2 agent with tools → `gptr_session` $text, $value, objects

Jev and Clef return one typed decision per input. Conversational models work in turns with tools and return a session.

SYSTEM 1: TYPED DECISIONS

gptr_login(`"typesafe")  # interactive R`
`e = ` **gptr_config**()`$egress`
`e$typesafe = "ack"`
gptr_config(`egress = e, .scope = "user"`)
Jev uses its own TypeSafe key and consent.

`abstracts = ` **c**(`"Randomized trial.",`
`  "Observational study."`)
`rct = ` **peter**(`"Is this an RCT?", abstracts,`
`  model = "jev"`)
Logical: TRUE when P(yes) ≥ 0.5 (the default threshold).

gptr_prob(`rct`)
`abstracts[rct]`; **sum**(`rct`)
P(yes); subset and count these typed answers.

peter(`"Which tissue?", "Hepatocyte",`
`  model = "jev",`
`  choices = c("liver", "lung", "other")`)
One label per input.

peter(`"How severe?",`
`  "Pain limits walking.", model = "jev",`
`  levels = c("low", "mid", "high")`)
A fractional score: the expected 0-based level.

peter(`"Is this an RCT?", abstracts,`
`  model = "jev", min_confidence = 0.8`)
Below the minimum (|2p−1| for yes/no), return NA or what `uncertain` says.

`fit_s |> ` **peter**(`"Did it succeed?",`
`  model = "jev"`)
Judge the live fit's last answer; add no turn.

SYSTEM 2: AGENTS

peter(`"Plan an mpg analysis.", mtcars,`
`  model = "gpt",`
`  .opts = list(thinking = "high")`)
Reasoning depends on the model; a suffix such as `gpt:high` works too.

Models and providers

```
"anthropic/claude-sonnet-5-5:high"
```
provider · model id · thinking

Aliases: `"sonnet"`, `"opus"`, `"haiku"`, `"gpt"`, `"flash"`, `"jev"`, `"claude_code"`, `"codex"`.

`anthropic/…`	`ANTHROPIC_API_KEY`
`openai/…`	`OPENAI_API_KEY`
`google/…`	`GEMINI_API_KEY`
`openrouter/…`	`OPENROUTER_API_KEY`
`azure/…`	`AZURE_OPENAI_API_KEY`, `AZURE_OPENAI_ENDPOINT`
`bedrock/…`	`AWS_BEARER_TOKEN_BEDROCK`, `AWS_REGION`
`ollama/…`	Local server; also lmstudio, vllm
`claude-cli`	Shell: `claude auth login`
`codex`	Shell: `codex login`
`jev`	`TYPESAFE_API_KEY`
`ollama/clef`	Installed model; Ollama 0.35.1+

```
gptr_models(provider = "ollama",
  refresh = TRUE)
```
Installed models on a running Ollama server.

```
gptr_register(gptr_provider("corp",
  "openai-completions",
  base_url = "https://HOST/v1",
  auth = "CORP_KEY", models = list(
    list(id = "MODEL",
      tool_call = TRUE))))
```
Template: replace HOST/MODEL; set CORP_KEY for calls. Same-name registration replaces it.

Permissions and trust

`plan`	Read only; next top-level call gets the plan
`manual`	Reads run; classified changes ask
`edits`	Also allows file edits in the project
`auto`	Critical actions and secret reads ask
`yolo`	Broad; critical and secret-flow guards remain

```
gptr_permissions(deny = "r(sh:rm *)")
```
Rules: deny beats ask beats allow.

```
gptr_rewind(s)
```
Page 1's s. Restore checkpoints; not references or external effects. Limits: `?gptr_rewind`.

Auto/yolo: computed `system()` needs an explicit `r(fn:system)` allow rule. Advisory, not a sandbox: `?gptr_security`.

Record, replay and export

R console, in order: use a new writable project and page 1's fake. Leave GPTR_REPLAY unset for this first recording.

`gptr_doc()` → `peter()` → `replay`

```
gptr_register(
  gptr_fake_provider(fake$script))
```
Reset/register the offline model.

```
writeLines("library(gptr)", "analysis.R")
gptr_doc("analysis.R")
s = peter("Fit mpg on wt.", mtcars,
  model = "fake/fake-1", mode = "auto")
gptr_doc(FALSE)
```
Create, bind, record a top-level call, then unbind.

```
gptr_source("analysis.R",
  replay = "replay")
gptr_blocks("analysis.R")
```
Replay; list fresh/stale/edited/undone blocks.

```
GPTR_REPLAY=replay Rscript analysis.R
```
POSIX shell (macOS/Linux), after recording. No model calls.

```
gptr_export("analysis.R", "plain.R")
```
Code runs; peter calls become comments. Needs fresh blocks and literal arguments.

```
gptr_export("analysis.R", "peter.R",
  format = "peter")
```
Calls run; code is commented. Live runs need the provider.

```
gptr_doc("report.Rmd")
```
Existing report: bind in setup before calls; rmarkdown/Pandoc renders it. Quarto needs its CLI.

```
gptr_doc("analysis.ipynb", sync = TRUE)
```
Existing recorded notebook: close before syncing.

Replay needs the code's inputs/packages; cached text needs .gptr/. Top-level `gptr_coach()` replays too.

Learn more

Articles, Harnesses and coaching, and the function reference.

```
peter$search("mcp server")
peter$help("gptr::gptr_mcp_add")
```
Search gptr's functions, members, tools and skills; read one's help, as Peter does.

Tools, files and artifacts

In a new writable project. Presets: minimal, standard, readonly (r can write), extended; select with `tools = "extended"`.

```
dir.create("R", showWarnings = FALSE)
peter$write("R/fit.R", "n=10\n")
peter$read("R/fit.R", limit = 40)
peter$grep("n=", path = "R")
peter$find("*.R"); peter$ls("R")
```
Create the fixture; read, search and list.

```
peter$edit("R/fit.R", list(list(
  oldText = "n=10", newText = "n=20")))
peter$write("notes.txt", "done\n")
```
Edit the fixture; write files atomically.

```
peter$sh(c("Rscript", "--version"))
peter$bg(c("Rscript", "--version"))
```
Run/background a program; Rscript uses this R.

```
peter$sql("SELECT cyl, AVG(mpg)
  FROM mtcars GROUP BY cyl", mtcars)
```
Data frames need DBI + duckdb. `peter$py()`: reticulate + Python.

```
apps = gptr_artifacts()
if (nrow(apps))
  gptr_artifacts(apps$id[[1]], open = TRUE)
```
After asking Peter to build a Shiny app: use its actual ID. Shiny required; scripts do not launch by default.

Skills, plugins and MCP

```
gptr_skills(); gptr_plugins()
```
List available skills and plugins.

```
peter("Review lm().", stats::lm,
  skills = "high-performance-r")
```
Bundled skill; needs page 1 model setup.

```
peter("Report units.", plugins = "./lab")
```
Requires trusted ./lab with plugin.json.

```
gptr_mcp_add("docs",
  url = "http://127.0.0.1:8000/mcp")
```
Your server's URL (or command for stdio); connects on use.

```
peter$mcp[["SERVER"]][["TOOL"]]()
```
Template: replace names and tool arguments.

```
mcp = gptr_mcp_serve()
as.character(mcp$config$claude_code)
```
httpuv/later/openssl needed. Paste into Claude Code; keep R idle (from Peter: `/mcp serve`). When done: `mcp$stop()`.

```
up = gptr_tool("shout", "Upper-case.",
  fun = function(x) toupper(x))
off = gptr_register(up)
```
Register a tool; off() removes it.

Teams, fan-outs, background

Needs page 1's model setup. Modes inherit; a team throws only if all children fail. Inspect statuses.

```
team = peter("Review mtcars.", mtcars,
  agents = list(
    stats = gptr_agent(system = "Stats"),
    code = gptr_agent(model = "gpt:low")),
  mode = "auto")
```
Named roles, each with its own model if set; `team$stats$text` reads one.

```
fan = peter("Count rows; gptr_return().",
  list(mtcars, iris), parallel = 2,
  mode = "auto")
```
One child per element; values in fan$value. `gptr_parallel()`: unrecorded calls; use separate envir values.

```
bg = peter("Count rows.", mtcars,
  background = TRUE, mode = "auto")
gptr_wait(bg)
```
Experimental; wait finishes the run.

```
gptr_usage(team, by = "agent")
```
Usage by session/agent/model/route.

Tailor and coach a harness

Page 1 model setup required. Coaching runs unattended R processes; most tasks need auto.

```
h = gptr_tailor(peter, mode = "auto",
  coaching = list(note = "Give units."))
```
A call's arguments win; its tools add.

```
h = gptr_tailor(h, rules = list(never =
  list(list(rule = "r(fn:stepAIC)",
    why = "No stepwise selection."))))
```
Hard never rule; require requests a missing step.

```
h("Describe mtcars.", mtcars)$text
```
Call like peter, pipes included.

```
x = c(1, 2, 3)
q = "Mean x; gptr_return() it."
h = gptr_coach(q, x, harness = h,
  check=function(r) isTRUE(r$value==2))
h = gptr_coach(q, x, harness = h, k = 2,
  strategy = "sequential")
```
Kept only if acceptance checks pass; a coached harness can be coached again, here in rounds that build on each other. Optional `coach = "claude_code"` needs its CLI login.

```
saveRDS(h, "h.rds")
h = gptr_tailor(readRDS("h.rds"))
```
Rebuild a saved harness; checks it and shows fixed code.

```
gptr_tailor(h, dir = "my-harness")
```
Write a new Markdown output folder; gptr never reads it back.